

EW – 1 Lab 6

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Part – A: ARDUINO DIGITAL

Arduino is an open-source platform based on an easy-to-use hardware and software. It has a user-friendly Development Environment (IDE). It was created keeping in mind the exposure of its users to electronics, thus is widely used to teach electronics. However, due to its versatility, it is extensively used as a prototyping platform for various large-scale electronics projects, and by electronics hobbyists. Another advantage is that it comes very cheap.

Task – 1: LED Sequential Control

AIM: Get 3 LEDs to turn on and off in a simple sequence.

Components Required:

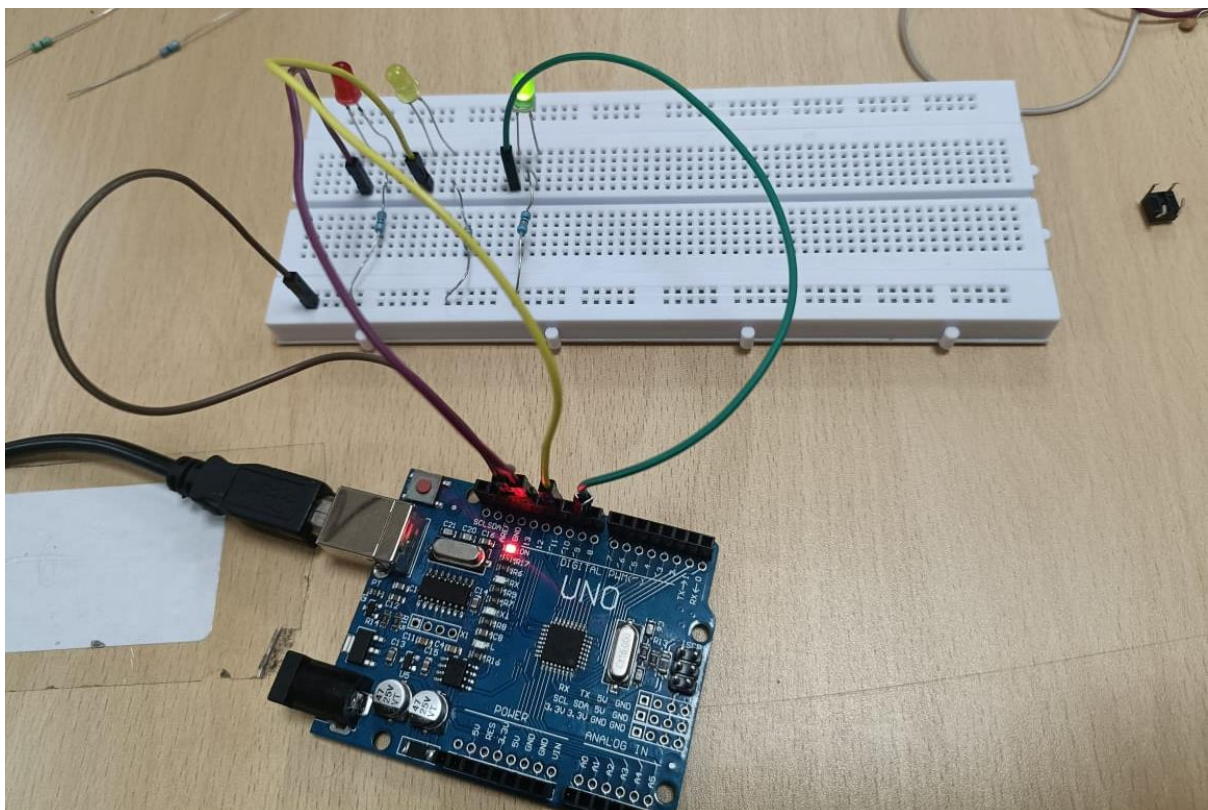
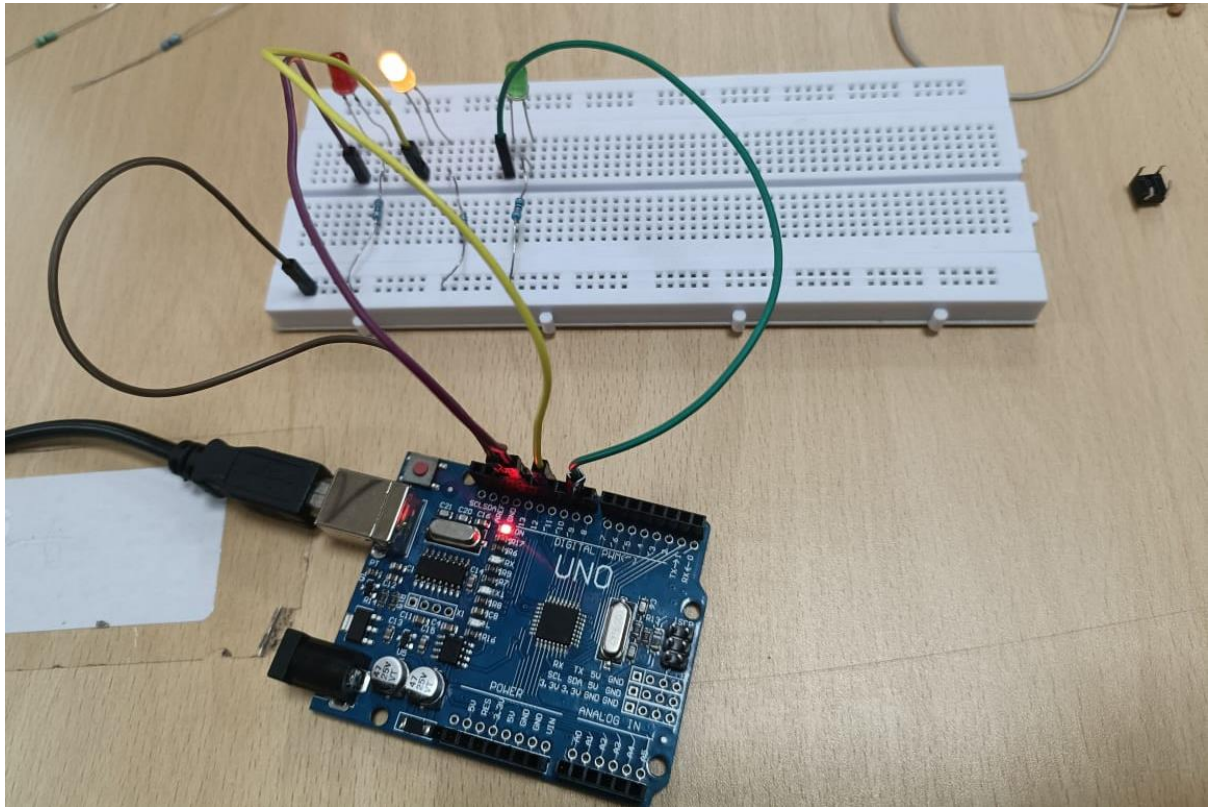
- Arduino Uno
- Breadboard
- LEDs x3
- 220 Ω resistors x3
- Jumper wires

Procedure:

- Plug each LED into the breadboard and connect their cathodes to GND of Arduino via separate resistors.
- Connect digital pins 9, 11 and 13 to the anodes of the LEDs.
- Upload the code for part A, task 1.
- Observe the output.

Results:

The LEDs glow turn-by-turn sequentially with a gap of half-a-second.



Observation:

The global variable 'state' holds the current state of the LEDs. It increments every 500 milliseconds and is reset when all the LEDs have been sequentially turned on and off once. This loop repeats forever.

Conclusion: LED sequential control using Arduino was successfully designed and tested.

Task – 2: RGB LED Control using Pushbutton

AIM: Control an RGB LED using a pushbutton.

Components Required:

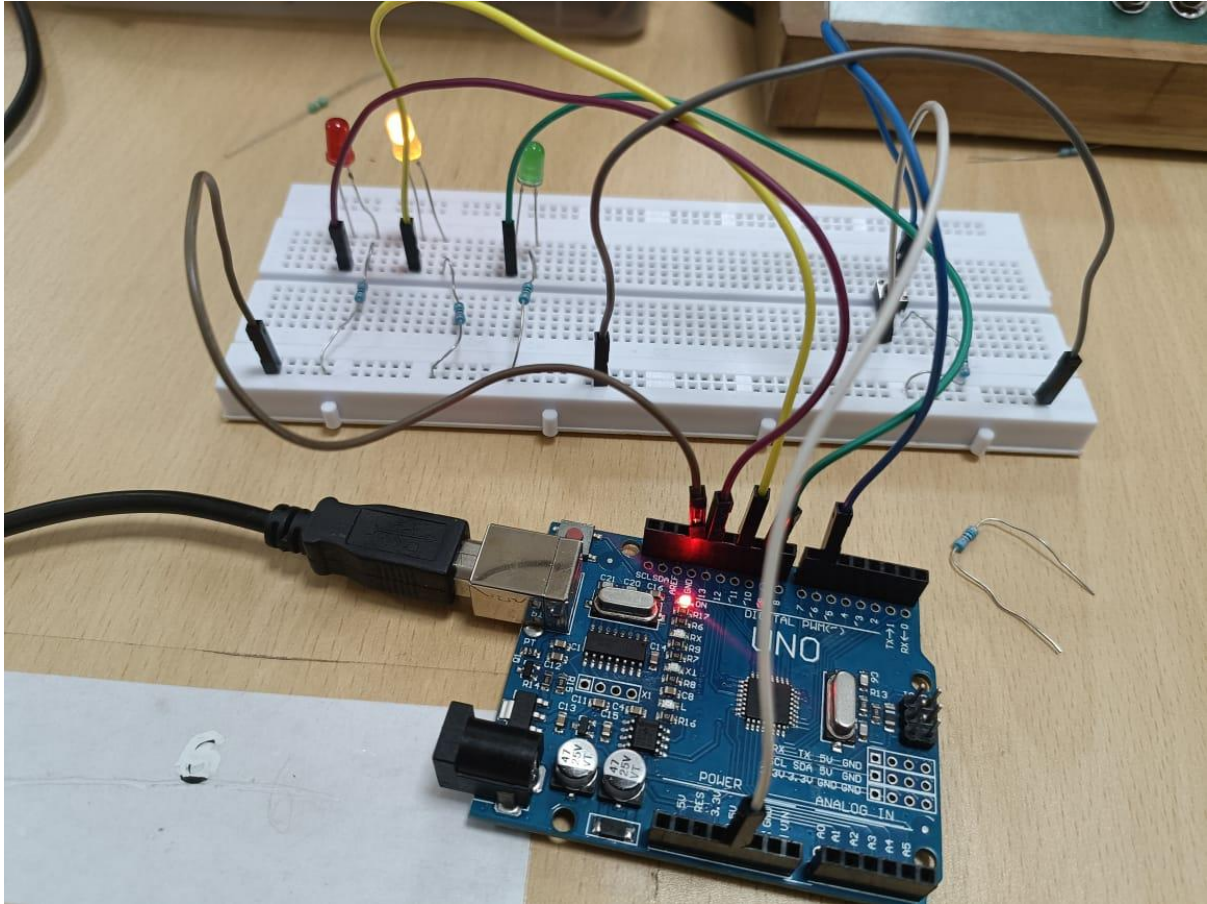
- Arduino Uno
- Breadboard
- RGB LED or different coloured LEDs x3 (the latter is used ahead)
- 220 Ω resistors x3
- 10 k Ω resistor
- Jumper wires

Procedure:

- Plug each LED into the breadboard and connect their cathodes to GND of Arduino via separate 220 Ω resistors.
- Connect digital pins 9, 11 and 13 to the anodes of the LEDs.
- Plug in the pushbutton and ground one terminal via the 10 k Ω resistor.
- Connect the other terminal to digital pin 6.
- Upload the code for part A, task 2.
- Observe the output as the button is pushed.

Results:

The LEDs glow turn-by-turn with every button press.



Observation:

The global variable 'state' holds the current state of the LEDs. It increments whenever the button is pressed and is reset when all the LEDs have been sequentially turned on and off once.

Conclusion: The LEDs/RGB LED were/was controlled successfully using a pushbutton.

Part – B: ARDUINO SENSING (Ultrasonic Sensor)

An ultrasonic sensor uses ultrasound (not audible to human beings) to calculate the distance between it and the nearest obstacle in its line of sight, by calculating the time between sound generation and echo detection. This is similar to SONAR technology and how bats search for their prey.

The ultrasonic sensor used here is the HC-SR04, with a maximum range of 4m. To get the distance, the time between the ultrasonic 'chirp' and the corresponding echo is noted, which is used in the usual formula for distance:

$$distance = 0.0343 \times time(\mu s)/2$$

Task – 1: Distance Measuring

AIM: Read the distance values from an ultrasonic sensor.

Components Required:

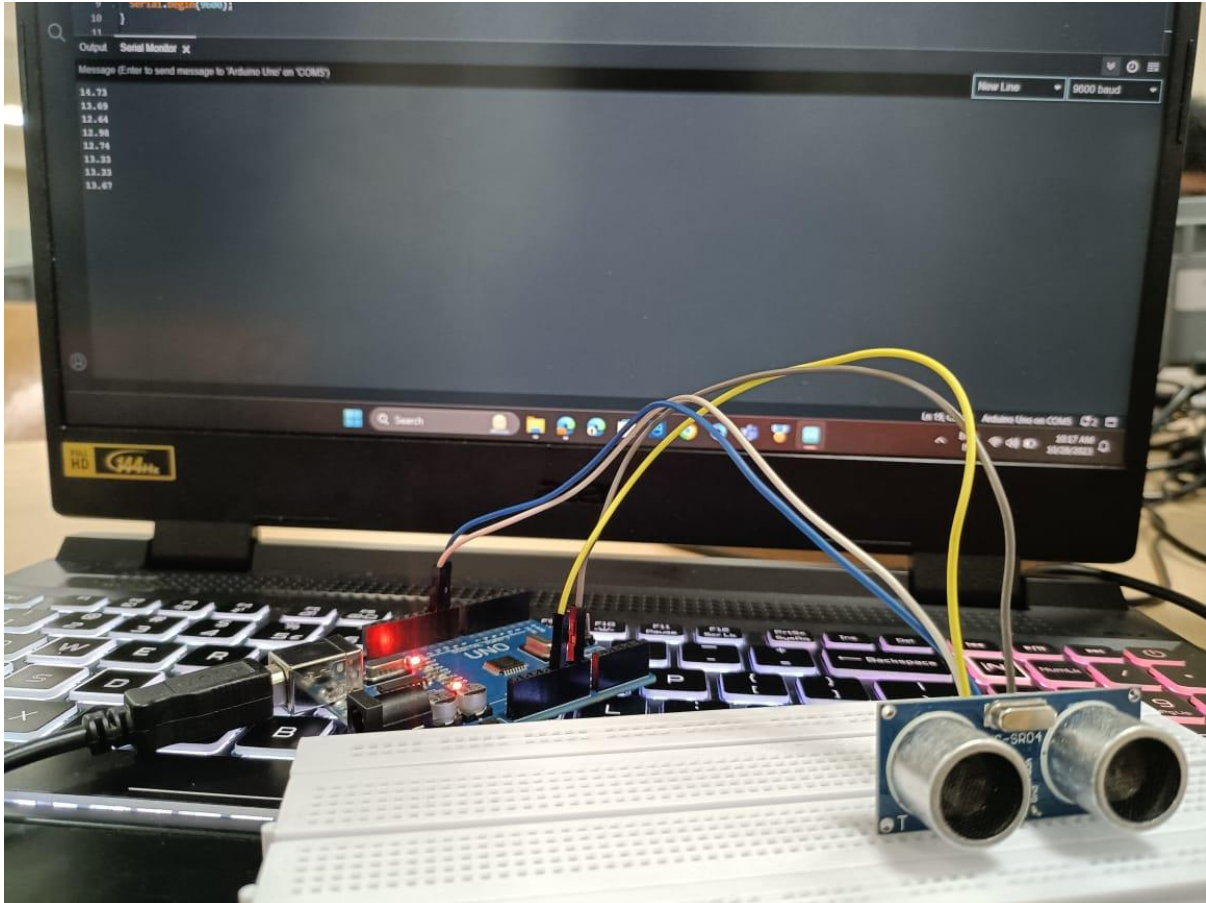
- Arduino Uno
- Breadboard
- HC-SR04 ultrasonic sensor
- Jumper wires

Procedure:

- Plug the sensor into the breadboard. Connect the power pins appropriately, providing 5V to V_{CC} .
- Connect the Trigger and Echo pins to digital pins 10 and 9 respectively.
- Upload the code for part B, task 1.
- Keep objects at some distance in front of the sensor and read the distance values off the serial monitor, in centimetres.
- Observe the values for different distances.

Results:

The distances in cm were read off the serial monitor, with minor errors, as shown:



Observation:

Marginal errors observed in the readings were due to the precision of the sensor itself, and electrical noise.

Conclusion: The ultrasonic sensor was successfully used to measure distances.

Task – 2: Proximity Alarm

AIM: Get an LED to glow if an object is in close proximity.

Components Required:

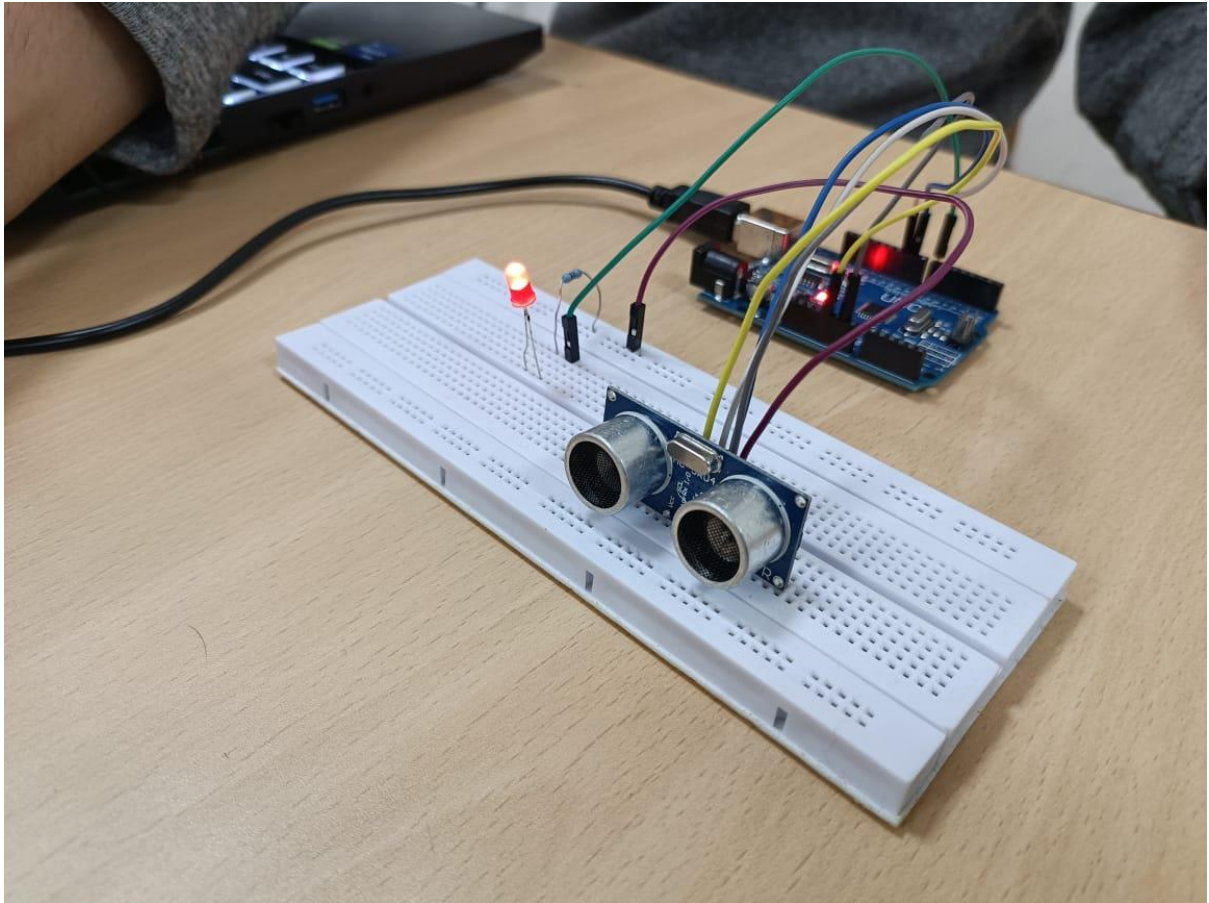
- Arduino Uno
- Breadboard
- HC-SR04 ultrasonic sensor
- Jumper wires
- LED
- 220 Ω resistor

Procedure:

- Plug the sensor into the breadboard. Connect the power pins appropriately, providing 5V to V_{CC} .
- Connect the Trigger and Echo pins to digital pins 10 and 9 respectively.
- Connect an LED with resistor on the breadboard. Provide input to it from digital pin 7.
- Upload the code for part B, task 2.
- Keep objects at some distance in front of the sensor and observe the LED as it glows and turns off.
- Note the distances for which the LED glows.

Results:

The LED glows for objects which are closer than 0.5 metres to the sensor, with marginal errors in measurements.



Observation:

Marginal errors observed in the readings were due to the precision of the sensor itself, and electrical noise.

Conclusion: The ultrasonic sensor was successfully used to build a proximity alarm.

Part – C: ARDUINO SENSING (Smoke Sensor)

A smoke sensor works by detecting smoke/soot particles in the air. The MQ-2 sensor is a metal oxide semiconductor (MOS) type smoke sensor. It uses a tin dioxide semiconductor material that adsorbs oxygen in the air, which reduces the density of electrons on the semiconductor and increases the resistance. When the sensor is powered on, the semiconductor heats to a high temperature, leading to greater adsorption and higher resistance, i.e., a low on the D0 pin. When a gas of high concentration is brought near to the sensor, it reacts with tin dioxide to reduce the adsorption of oxygen, thus decreasing resistance by increasing electron density. That is,

- Greater the gas concentration, greater the output voltage (lower the resistance of the semiconductor).
- Lower the gas concentration, lower the output voltage (higher the oxygen adsorption and higher the resistance).

The MQ-2 sensor can detect concentrations in the range of 200-10000 ppm.

AIM: Read the smoke sensor analogue output voltage and glow an LED if the smoke level exceeds a certain threshold.

Components Required:

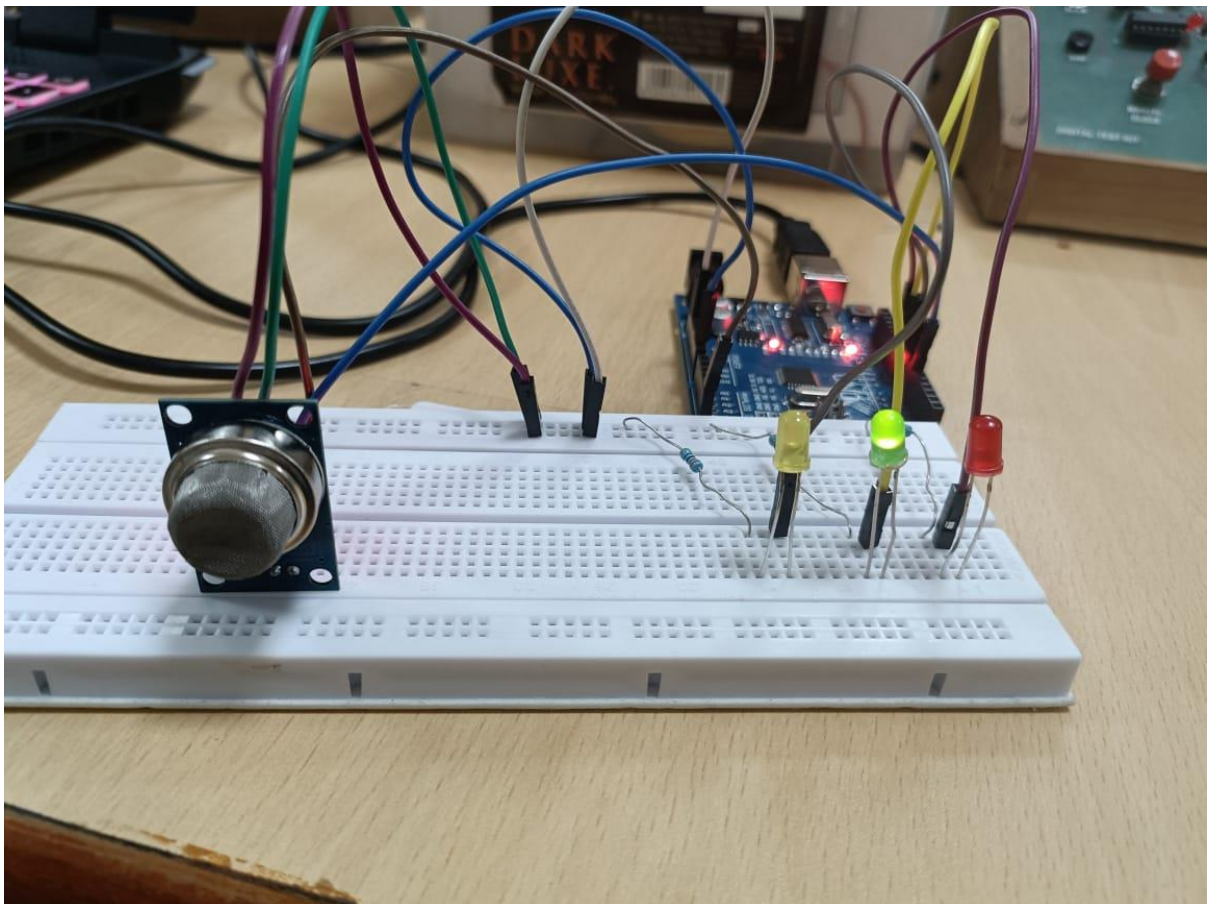
- Arduino Uno
- Breadboard
- MQ-2 smoke sensor
- Jumper wires
- LEDs
- 220 Ω resistors
- Incense stick and lighter

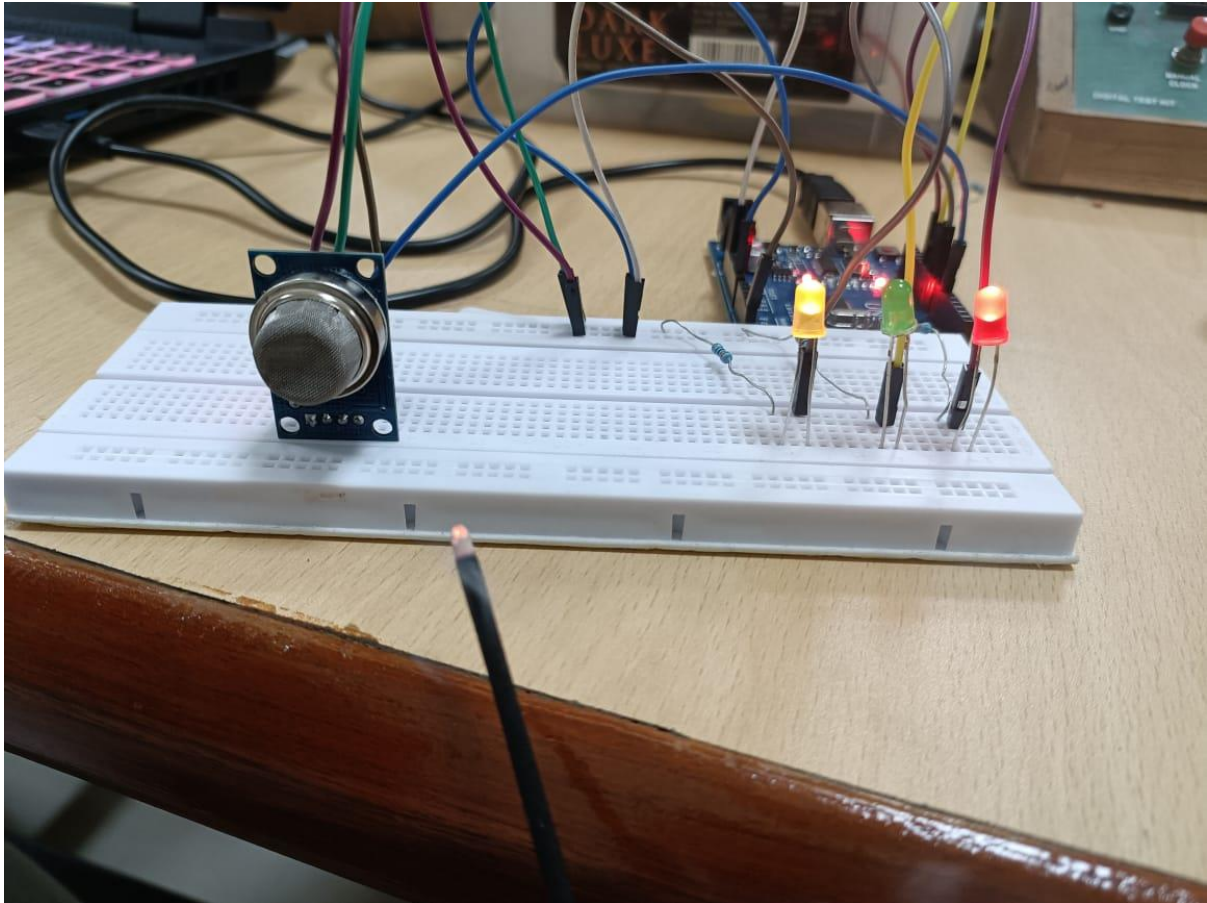
Procedure:

- Plug the sensor into the breadboard. Connect the power pins appropriately, providing 5V to V_{CC} .
- Connect the D0 and A0 pins to pins 6 and A5 respectively.
- Connect a red and a green LED with resistors on the breadboard. Provide input to them from digital pins 12 and 10 respectively.
- Upload the code for part C.
- Notice the LEDs when there is no smoke.
- Now, bring a burning incense stick near the sensor and observe the LEDs.
- Take note of the analogue voltage readings on the serial monitor in both the cases.

Results:

The following images show the results in both the cases:





Observation:

The green LED glows when the gas concentration is less than 200 ppm. On increased concentration, the red LED glows and green LED turns off.

On the serial monitor, D0 is 1 whenever A0 exceeds 512, which is the mean value of all analogue step voltages.

Conclusion: The smoke sensor successfully detected excess smoke concentration.